

IAL Physics Unit 6 (WPH16/01) — Complete A* Study Note

Practical Skills in Physics II

Built entirely from analysis of 17 past papers and their mark schemes, June 2019 → June 2025.

Unit 6 is not a content paper — it is a **skills** paper, and the same eight skills are tested in every session wearing different physics costumes. That is the good news: **the answer banks are finite and the mark schemes repeat almost verbatim year after year**. Learn them and this becomes the most predictable paper in the whole IAL.

PART A — THE PAPER AND HOW IT IS MARKED

1. The paper

Code	WPH16/01 — Unit 6: Practical Skills in Physics II
Time	1 hour 20 minutes
Marks	50
Questions	4, worth roughly 6–18 marks each
You must have	Scientific calculator, ruler
Formula sheet	Printed at the end of the booklet

The one rubric instruction that is a marking rule:

"Show all your working out in calculations and include units where appropriate."

Pace: 50 marks in 80 minutes = **1.6 minutes per mark** — the most generous rate of any IAL physics paper. Use it. The marks here are won by care, not speed.

Why Unit 6 matters disproportionately: it carries **60 UMS**, and the A* is decided by **≥ 90**

% of the UMS across Units 4, 5 and 6 combined ($\geq 270/300$). Unit 6 is a fifth of that calculation — and it is by far the easiest fifth to secure, because the mark schemes are so repetitive.

2. The structure — identical in every paper

Q	Typical marks	What it almost always is
1	6–9	A short practical context: safety, completing a circuit, criticising a graph, a small uncertainty calculation
2	6–9	"Devise a method" or "criticise / improve this plan" — usually a 6-mark structured description
3	14–18	The big one. A relationship of the form $y = Ae^{(-Bx)}$ or $y = ax^n$, linearised with ln or log , then: table $\rightarrow$ plot $\rightarrow$ gradient $\rightarrow$ determine a constant $\rightarrow$ use it
4	14–16	The measurement question. Measure an object with vernier/micrometer, compute a derived quantity, combine percentage uncertainties , then deduce whether it matches an accepted value

Roughly 30 of the 50 marks are in Questions 3 and 4, and both follow a fixed recipe. Master those two recipes and the grade is essentially secured.

3. The marking rules that decide marks

(a) Units.

*"A separate mark is not usually given for a unit but **a missing or incorrect unit will normally mean that the final calculation mark will not be awarded.**" "This does not apply in 'show that' questions or... where the units to be used have been given." "The mark scheme will indicate if no unit error is to be applied by **placing brackets around the unit.**"*

Write a unit on every final answer line. The one systematic exception is a **gradient of a log graph**, which the schemes mark as *"negative, no unit"* or *"Allow unit of mm^{-1} "* — see §12.

(b) Significant figures — Unit 6 is stricter than the theory papers.

"In questions assessing practical skills, a specific number of significant figures will

be required e.g. determining a constant from the gradient of a graph or in uncertainty calculations. The MS will clearly identify the number."

The conventions the papers actually use:

Gradient of a graph	2 or 3 s.f.
Percentage uncertainty	1 or 2 s.f.
Mean of repeated readings	SAME d.p. as the raw data (or as specified)
Absolute uncertainty	d.p. CONSISTENT with the mean
Processed table values (ln, log)	consistent, usually 2-3 d.p. or 3-4 s.f.

(c) $g = 10$ is penalised.

"The use of $g = 10 \text{ m s}^{-2}$ instead of 9.81 will be **penalised by one mark** (but not more than once per clip). Accept 9.8."

(d) "Show that" questions and reverse working.

"If a 'show that' question is worth **2 marks**, then **both marks** will be available for a reverse working. If the question is worth **3 marks**, then **only 2 marks** will be available."
"Too few significant figures will mean that the final mark cannot be awarded in 'show that' questions where **one more significant figure than the value in the question is needed**."

So: work forwards, and quote one more significant figure than the target. "Show that it is about 2%" → give **2.1 %**.

(e) "Use of the formula" means physically correct substitution.

"**use of the formula** means that the candidate demonstrates **substitution of physically correct values**, although there may be conversion errors e.g. power of 10 error."

A power-of-ten slip still earns the method mark — provided the substitution is visible. Always write it.

(f) Underlining and bold in the mark scheme.

"Alternative correct wording should be credited **unless the MS has specified specific words that must be present**. Such words will be indicated by underlining, e.g. 'resonance'." "**Bold lower case** will be used for emphasis e.g. '**and**' when two pieces of information are needed for 1 mark."

4. ⚠️ Dependent marks — the structural feature of this paper

More marks in Unit 6 are dependent than in any other IAL physics paper. The schemes

say "MP2 dependent on MP1" constantly. Three examples, all verbatim:

"Repeat at different orientations and calculate a mean (1) — To reduce (the effect of) random error (1). **MP2 dependent on MP1. Allow 1 mark only for "repeat and calculate a mean to reduce (the effect of) random errors".**"

" $\ln V = \ln A - Bw$ (1) — Compares to $y = c + mx$ where the gradient is $-B$ (1). **MP2 dependent on MP1.**"

"Correct value of relevant limit shown (1) — Conclusion based on comparison of relevant limit and accepted value (1). **MP2 dependent MP1.**"

The rule to work by: never give a conclusion without the statement it depends on.

Write the *what* and then the *why* as two separate sentences.

PART B — MEASUREMENT

5. Instruments and their resolutions — learn this table

Instrument	Resolution	Uncertainty in a single reading	Use for
Metre rule	1 mm	± 0.5 mm, or ± 1 mm if both ends are judged	lengths $> \sim 10$ cm
30 cm ruler	1 mm	± 0.5 mm	medium lengths
Vernier callipers	0.1 mm (some 0.02 mm)	± 0.05 mm	diameters, small lengths, internal/external
Micrometer screw gauge	0.01 mm	± 0.005 mm	wire and sheet thicknesses
Travelling microscope	0.01 mm	± 0.005 mm	very small separations
Digital balance	0.01 g or 0.001 g	half the resolution	masses
Stopwatch (hand)	0.01 s, but reaction time ≈ 0.2 – 0.3 s	± 0.1 s or larger	long times only

Light gate / data logger	~0.001 s	very small	short times
Thermometer (mercury)	1 °C or 0.5 °C	half the resolution	
Temperature probe + logger	0.1 °C or better	smaller	
Protractor	1°	±0.5°	angles
Ammeter / voltmeter	depends on the display	half the last digit	

⚠ For a digital display, the uncertainty is normally taken as $\pm$ the last digit or $\pm$ half the last digit — the schemes accept either provided you state which.

6. ⚠ "Explain why this instrument was appropriate" — the two-mark recipe

This appears in most papers and the mark scheme is always the same shape (June 2024 Q4a(ii), June 2025 Q3a):

*"The micrometer screw gauge has a **resolution of 0.01 mm** — Or the measurement will have an **uncertainty of 0.005 mm (1)**" "So the **percentage uncertainty is 0.35 % which is small (1)**" "Do not accept accuracy/precision, or implied in calculation."*

So the answer has exactly two parts, and both must be explicit:

1. STATE the resolution (or half-resolution) as a NUMBER WITH A UNIT
2. CALCULATE the percentage uncertainty and COMMENT that it is small

$$\%U = 0.005 \text{ mm} \div 1.21 \text{ mm} \times 100 = 0.4 \%, \text{ which is small}$$

⚠ Three traps named by the scheme:

- Saying the instrument is "**accurate**" or "**precise**" — **not accepted**
- Doing the calculation **without stating the resolution first** — "*or implied in calculation*" is rejected
- Omitting the comment that the value **is small**

7. ⚠ "Describe a technique to get an accurate value" — the two-mark pairs

Another near-universal question. The mark scheme gives exactly two acceptable pairs, and both are dependent (2023 Jan Q4a(i), Oct 2024 Q4a, June 2024 Q4a(i), June 2025 Q4b(i)):

EITHER *"Repeat at different orientations (or at different places along the object) and calculate a mean (1)" "To reduce (the effect of) random error (1) — MP2 dependent on MP1. Do not allow eliminate/avoid."*

OR *"Check and correct for zero error (1)" "To eliminate/reduce systematic error (1) — MP2 dependent on MP1. Do not allow system/systemic."*

"ONE mark only for 'repeat and calculate a mean to reduce random errors!'" "ONE mark only for 'check for zero error to eliminate systematic error!'"

⚠ The word that earns the second mark is **"at different orientations"** or **"at different places"**. Plain "repeat and average" is a single mark. And note the Oct 2024 variant: for a metre rule the scheme says *"Repeat at different **places**. Do not accept orientation"* — match the word to the object.

⚠ **Vocabulary:** *reduce* random error (never "eliminate"); *eliminate* systematic error. And write **systematic** — *"do not allow system/systemic."*

8. Timing oscillations — the standard answer bank

"Explain why timing multiple oscillations improves the measurement of T" (3 marks, June 2025 Q4a(i)):

*"The **measurement (of time) will be larger** (1) — Do not accept 'time period' or 'T will be larger'" "But the **uncertainty will be constant** (1)" "So the **percentage uncertainty will be reduced** (1)"*

⚠ Say **"the measured time"**, not **"the time period"**. T itself does not change.

"Describe techniques for timing oscillations" (2–3 marks):

- Use a TIMING MARKER at the CENTRE of the oscillation
(the object moves fastest there, so the timing is most precise)
- START TIMING AFTER SEVERAL OSCILLATIONS have completed
- Use a SMALL INITIAL DISPLACEMENT
- Time MULTIPLE OSCILLATIONS and divide by the number
- REPEAT and calculate a MEAN

These five points, drawn verbatim from the June 2020 and June 2025 schemes, cover every version of this question.

PART C — UNCERTAINTY

The single highest-yield topic on the paper: 10–15 marks a session.

9. The core definitions and formulae

Absolute uncertainty from a SET OF REPEATS:

$$\text{uncertainty} = \text{HALF the RANGE} = (\text{max} - \text{min}) / 2$$

or the reading FURTHEST FROM THE MEAN

Absolute uncertainty in a SINGLE reading = half the resolution

(or the full resolution if two judgements are made, e.g. both ends of a length)

$$\text{Percentage uncertainty } \%U = (\text{absolute uncertainty} \div \text{value}) \times 100$$

⚠ The most expensive single ruling on the paper (June 2025 Q4a(iv)):

"Use of full range scores 0."

Always halve the range. This alone costs candidates two marks every session.

Worked, from that scheme:

10T values: 6.88, 6.93, 6.84, 6.96 s

Mean 10T = $27.61/4 = 6.90$ s → Mean T = 0.690 s ("given to 2 d.p. only" →

Half range = $(6.96 - 6.84)/2 = 0.06$ s

$\%U = 0.06/6.90 \times 100 = 0.87\% = 0.9\%$ ("given to 1 or 2 s.f.")

Note that the **percentage uncertainty is the same whether you use 10T or T** — the factor of 10 cancels. Use the raw times.

10. ⚠ Combining uncertainties — the four rules

ADDING or SUBTRACTING → ADD the ABSOLUTE uncertainties

MULTIPLYING or DIVIDING → ADD the PERCENTAGE uncertainties

RAISING TO A POWER n → MULTIPLY the percentage uncertainty by n

MULTIPLYING BY A CONSTANT → percentage uncertainty UNCHANGED

⚠ **The rule students miss is the first one.** For $A = \pi/4 (s^2 - d^2)$, the 2023 Jan scheme awards a mark for **"Addition of uncertainties in s^2 and d^2 "** — the **absolute** uncertainties in the two squares are added, **not** the percentages. Constants like $\pi/4$ contribute nothing.

Worked — the classic power question (June 2025 Q4b(ii)):

Show that the %U in $D^2/2d^2$ is about 2 %, $D = 59.4 \pm 0.2$ mm, $d = 13.9 \pm 0.1$ mm

$$\%U \text{ in } D = 0.2/59.4 \times 100 = 0.34 \%$$

$$\%U \text{ in } d = 0.1/13.9 \times 100 = 0.72 \%$$

$$\%U \text{ in } D^2/2d^2 = 2 \times 0.34 \% + 2 \times 0.72 \% = 2.1 \% \quad \checkmark \quad (\text{"given to minimum 2 s.f."})$$

The three marks are: (1) computing both %Us, (2) doubling both, (3) the answer to ≥ 2 s.f. The constant 2 in the denominator is ignored.

Worked — a four-quantity formula (Oct 2024 Q4b(iii)):

$$E = 16\pi^2 MD^3 / (ab^3 T^2)$$

$$\begin{aligned} \%U \text{ in } E &= \%U \text{ in } M + 3(\%U \text{ in } D) + \%U \text{ in } a + 3(\%U \text{ in } b) + 2(\%U \text{ in } T) \\ &= \dots + 3(0.125) + 0.200 + 3(0.324) + 2(1.61) \\ &= 4.77 \% \end{aligned}$$

*"Uses $3 \times \%U$ in D **or** $3 \times \%U$ in b **or** $2 \times \%U$ in T (1)" — the powers are a mark in themselves.*

11. The alternative route — max/min

Every uncertainty mark scheme offers a second method, worth the same marks:

1. Compute the MAXIMUM value using the extreme values that maximise it
2. Compute the MINIMUM value likewise
3. Uncertainty = (max - min) / 2 $\leftarrow$ HALF the range again
4. Percentage uncertainty = that $\div$ the central value $\times 100$

Use this when the formula has terms that combine awkwardly (sums inside powers, or a subtraction inside a product). The June 2024 volume question and the 2023 Jan area question both credit it fully.

⚠ **Be careful which extreme goes where.** To maximise $(s^2 - d^2)$ you take **s at its maximum and d at its minimum.**

12. ⚠ "Deduce whether..." — the two-mark conclusion recipe

Every paper ends Question 4 with this, and the mark scheme is always the same two dependent points:

"Correct value of relevant limit shown (1) — e.c.f." "Conclusion based on comparison of relevant limit and accepted value (1) — MP2 dependent MP1."

Method A — the limit method (preferred):

1. Calculate the UPPER and/or LOWER LIMIT of your value:
$$\text{limit} = \text{value} \times (1 \pm \%U/100)$$
2. STATE whether the accepted value lies INSIDE that range
3. CONCLUDE in a sentence

Worked (June 2025 Q4b(iii)): $v = 0.276 \pm 6\%$, steel is 0.265.

Lower limit = $0.276 \times 0.94 = 0.259$; Upper limit = $0.276 \times 1.06 = 0.293$
0.265 lies between 0.259 and 0.293
→ the value is consistent with steel, so the spring MAY BE made of steel

Method B — the percentage difference method:

$$\%D = |\text{experimental} - \text{accepted}| \div \text{accepted} \times 100$$

Compare %D with the %U:

$\%D < \%U \rightarrow$ consistent / agrees within experimental uncertainty
 $\%D > \%U \rightarrow$ NOT consistent

Worked (June 2024 Q4c): "%D = 12 % (1) — As %D is **greater than 3 %** the protractor **may not be** made of Perspex (1)."

⚠ **Phrase the conclusion tentatively.** The schemes use "may be" and "may not be" — an experiment consistent with a value does not prove it.

13. Uncertainty in a gradient

For the best-fit line, draw the **steepest and shallowest lines that still pass through all the error bars**, then

$$\text{uncertainty in gradient} = (\text{steepest gradient} - \text{shallowest gradient}) / 2$$

— half the range again. The same idea gives the uncertainty in an intercept.

PART D — ERRORS

14. Random versus systematic

	Random error	Systematic error
Effect	Readings scatter either side of the true value	All readings shifted in the same direction by the same amount
On a graph	Points scatter about the line	Line has the wrong intercept (parallel to the correct one)
Causes	Judging a reading, reaction time, fluctuating conditions	Zero error , wrongly calibrated instrument, parallax done consistently, background count
Reduced by	REPEATING and averaging ; using more precise instruments	Cannot be reduced by repeating
Removed by	—	Checking and correcting for zero error ; subtracting a known offset; calibration

⚠ **Vocabulary the schemes enforce:** you **reduce** random error (never "eliminate"); you **eliminate** or **remove** systematic error. Write **systematic** in full.

The background-count answer (June 2019 Q4a):

*"Background count rate is a **predictable/constant value** (1)" "(hence) is a **systematic error** (which can be **subtracted**) (1)"*

Parallax: read a scale **perpendicularly**, with the eye level with the marker; use a **mirror scale** or a **set square**. Parallax done inconsistently is random; done consistently it is systematic.

15. Accuracy, precision, resolution — the three words

ACCURACY — how close a reading is to the TRUE value (freedom from systematic error)
PRECISION — how closely REPEATED readings agree with each other (small random scatter)
RESOLUTION — the SMALLEST CHANGE the instrument can detect

A set of readings can be precise but inaccurate (tightly clustered around the wrong value, from a zero error). **Never use "accurate" to justify an instrument choice** — use **resolution** (§6).

16. Data loggers — the three-mark answer

(June 2025 Q2b):

*"Data logger **records pressure and temperature simultaneously** (1)" "Temperature probe has a **better resolution** — Or **smaller uncertainty** — Or **reduces parallax error** (1)" "So the **readings will be closer to the true values** (1) — **MP3 dependent on MP1 or MP2**"*

Other creditable advantages (from other papers): records at a **high rate**, so it catches **rapidly changing** quantities; removes **human reaction time**; can record for a long time unattended; produces the graph directly.

⚠ That particular scheme says *"Ignore references to sampling rate and reaction time"* — **because the question asked specifically about accuracy**. Read what is being asked: *accuracy* → resolution, parallax, true values; *reliability* → more readings, repeats.

PART E — GRAPHS

Question 3 is built on graph skills and is worth 14–18 marks. Every one of the following is a separate mark.

17. ⚠ The plotting checklist — quoted from the general guidance

*"6.1 A mark given for axes requires **both axes to be labelled with quantities and units** and **drawn the correct way round**." "6.3 A mark given for choosing a scale requires that the chosen scale allows all points to be plotted, **spreads plotted points over more than half of each axis**, and is **not an awkward scale e.g. multiples of 3, 7**." "6.4 Points should be plotted **to within 1 mm**. Check the **two points furthest from the best line**. If both OK award the mark." "For a line mark there must be a **thin continuous line** which is the **best-fit line** for the candidate's results."*

The five marks on a plotting question, and how to secure each:

Mark	What to do
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Processed values	Fill the extra column; keep consistent d.p. or s.f. (usually 2–3 d.p. for ln values)
Axes	Label quantity AND unit on both. For a log axis write $\ln (V / mV)$ — the unit goes inside
Scale	Points must span more than half of each axis. Use 1, 2, 5 (or 10, 20, 50) per square. Never 3 or 7
Plotting	Every point to within 1 mm . Use sharp pencil crosses (x) , not blobs
Best-fit line	A thin continuous line, balanced with points either side. Not dot-to-dot, not forced through the origin

⚠ For a log axis you do **NOT** have to start at zero. A false origin is expected and is the usual way to spread the points over more than half the axis.

18. ⚠ Calculating a gradient — the three marks

(June 2025 Q3c(iii)):

"Uses large triangle to calculate gradient (1) — [data from table must be on best fit line]" "Value of gradient in range (1)" "Calculated gradient given to 2 or 3 s.f., negative, no unit (1) — Allow unit of mm^{-1} ." "Bald answer can score MP2 only."

The routine:

1. Draw a triangle spanning AT LEAST HALF the line — mark it ON THE GRAPH
2. Read BOTH coordinates from the LINE, not from the data table (unless a table point lies exactly on the line)
3. Write the subtraction out: $(y_2 - y_1) / (x_2 - x_1) = \dots$
4. Give 2 or 3 s.f., WITH THE SIGN, and with a unit if one is meaningful

⚠ "Bald answer scores MP2 only" — a gradient value with no visible triangle and no subtraction loses two of the three marks. **Show the triangle and the arithmetic.**

19. ⚠ Linearising a relationship — the two-mark explanation

This is the hinge of Question 3, and both marks are dependent.

Exponential relationships → take natural logs

$$V = A e^{(-Bw)}$$

$$\ln V = \ln A - Bw$$

Compare with $y = mx + c$:

$$y = \ln V, \quad x = w, \quad \text{gradient } m = -B, \quad \text{intercept } c = \ln A$$

The mark scheme (June 2025 Q3c(i)):

" $\ln V = \ln A - Bw$ (1)" **"Compares to $y = c + mx$ where the gradient is $-B$ (which is constant) (1) — MP2 dependent on MP1"**

So write TWO lines: the rearranged log equation, then the explicit comparison to $y = mx + c$ naming what the gradient and intercept represent.

Power relationships → take logs (base 10 or natural)

$$T = k I^n$$

$$\log T = n \log I + \log k$$

Compare with $y = mx + c$: gradient = n , intercept = $\log k \rightarrow k = 10^{(\text{intercept})}$

(June 2020 Q3b(ii)): " $\log T = n \log I$ is in the form of $y = mx (+ c)$ (1) — (hence if the relationship is valid) it will be a **straight line (through the origin) with a gradient of n (1).**"

Which to use

Form	Plot	Gradient	Intercept
$y = Ae^{(kx)}$	ln y against x	k	$\ln A$
$y = Ax^n$	log y against log x	n	$\log A$
$y = mx + c$	y against x	m	c
$y = k/x$	y against $1/x$	k	0
$y = k/x^3$	y against $1/x^3$	k	0
$C = C_0 e^{(-\lambda x)}$	ln C against x	$-\lambda$	$\ln C_0$
$T = 2\pi\sqrt{1/g}$	T^2 against l	$4\pi^2/g$	0

⚠ **"Determine a value for the constant" then means: gradient → rearrange → the constant.** And "how would you check the relationship is valid?" is answered by **"a straight line (through the origin) would confirm it".**

20. Curves and maxima

Some questions need a **smooth curve**, not a line. The scheme (June 2019 Q3a) asks for "**smooth curve drawn through points**" and "**maximum value on best fit line between 820 and 835 nm by eye**". **Draw the curve first, then read the peak off the curve, not off the highest data point.**

PART F — PLANNING AND DEVISING A METHOD

Question 2 is usually "*Devise a method...*" worth 6 marks, or "*criticise this plan*".

21. Δ The six-point template — every mark scheme fits it

From the June 2025 Q2a scheme (estimating absolute zero) and the June 2019 Q2b scheme (determining a constant k), the six creditable areas are always:

1. APPARATUS — name any ADDITIONAL equipment, specifically
"a hot plate / Bunsen burner / water bath", "a beaker of ice",
"a metre rule", "an ammeter", "vernier callipers"
2. HOW TO VARY the independent variable
"place the flask in a beaker of ice", "add glass slides one at a time"
3. WHAT TO MEASURE, AND WITH WHAT
"record the temperature from the thermometer AND the
corresponding pressure from the pressure gauge"
4. A TECHNIQUE or CONTROL that improves the measurement
"heat gradually", "stir the water", "read the scale perpendicularly",
"ensure source and detector are in line", "subtract the background count"
5. RANGE AND NUMBER OF READINGS
"record at a MINIMUM of 5 different values" OR
"at intervals of 20 °C or less" OR
"repeat until the count rate reaches background"
 Δ "Must be for DIFFERENT values, not to repeat and calculate a mean"
6. THE GRAPH — what to plot AND how to use it
"plot pressure against temperature and EXTRAPOLATE the line to pressure = 0
"plot V against $1/d^3$ and determine k from the gradient = kNI "

Δ Point 6 carries the most weight and is the one most often half-answered. "Plot a

graph" alone is not a mark — you must say **what against what** and **what you do with it** (gradient, intercept, extrapolation).

The scheme also notes: "Accept a **clearly labelled sketch graph**" — so if you are short of words, draw it and label the axes.

22. Choosing the graph — the alternatives are all credited

The June 2019 scheme shows how flexible this is. For $V = kNI/d^3$ it lists:

y	x	gradient	k from
V	$1/d^3$	kNI	gradient $\div$ NI
V/I	$1/d^3$	kN	gradient $\div$ N
V/N	$1/d^3$	kI	gradient $\div$ I
V/NI	$1/d^3$	k	= gradient
log V	log d	—	intercept = log(kNI)

Any correct linearisation scores. Pick the one that puts the constant you want straight into the gradient.

23. "State the main source of uncertainty"

The mark scheme's accepted answers (June 2019 Q2b(iv)) show what a good answer looks like — it names the measurement **AND** says why:

*"The secondary coil is **not parallel** to the primary coil or changes alignment"*

*"Measurement of d, as it is **difficult to align the ruler**" "Measurement of d, as **it is cubed hence the %U is tripled**"*

The third is the model answer. Identify the quantity with the **largest percentage uncertainty**, and if it is raised to a power, say so — *"because it is cubed, its percentage uncertainty is tripled."*

24. Safety

Ignore generic answers. The June 2025 scheme says outright: **"Ignore references to switching off power supply, general safety issues and PPE."** Goggles and "be careful" score nothing.

Give a hazard specific to the apparatus:

Context	Specific precaution
Electrolytic capacitor	Do not exceed the working p.d. · connect with the correct polarity · discharge fully before handling
Radioactive source	Handle with tongs / at a distance · use shielding · minimise exposure time · store in a lead-lined box
Hot water / heating	Use tongs or heatproof gloves ; clamp the beaker; keep the flex away from the hotplate
Falling masses	Place a cushion or tray beneath; keep feet clear
Stretched wire / spring	Safety goggles — the wire may snap and whip back
Laser	Do not look into the beam; no reflective surfaces; beam below eye level
Mains apparatus	Check insulation; keep water away from the supply

The June 2020 radioactivity scheme credits simply: ***"Sensible comment on safety based on distance, shielding or time."***

PART G — CRITICISING AND IMPROVING

25. ⚠ "Criticise the student's graph" — the answer bank

(June 2025 Q1b, 3 marks — and note what is explicitly excluded):

*"There are **not enough data points** (1)" "The **range of data is too small** — Or the **scales on the graph are too small** (1)" "The **spread of data about the best fit line is not reasonable** — Or the **best fit line may not be linear** (1). Allow reference to data points indicating a curve."*

"Ignore references to significant figures, repeats, inconsistent intervals." "Do not allow reference to an anomaly/outlier." "Ignore reference to origin."

And from June 2019 Q3b, for a curve:

"No repeats shown · Not enough readings around the maximum value · Best fit line

is uncertain as it is a curve"

The full bank to choose from — pick the ones that fit the graph in front of you:

- Not enough data points (fewer than 6)
- Range of the independent variable too small
- Scales too small / points span less than half the axis
- Axes not labelled with units
- No error bars
- Points scattered too widely about the line — the line is uncertain
- The line may not be linear / the points suggest a curve
- Not enough readings in the region of interest (near a peak or turning point)
- No repeats shown

⚠ **Read the actual graph.** These are not generic — the scheme rewards the ones that genuinely apply, and explicitly ignores several plausible-sounding ones.

26. "Criticise the plan" / "explain how this could be improved"

Apply the §21 template in reverse — find which of the six elements is **missing or inadequate**:

- No control of a relevant variable
- Range too small or too few values
- No repeats
- Instrument resolution too coarse for the quantity
- No named measuring instrument
- No mention of how the graph would be used
- A safety omission specific to the apparatus
- A source of systematic error not addressed (zero error, background count)

27. "Explain one variable she should control"

The mark scheme wants the variable AND why it matters (June 2025 Q3b, 2 marks):

*"The **distance between the filament bulb and the slides/solar cell** (1)" "As the **intensity of light varies as an inverse square law** (1)" Or: "The **background light** (1) — as any variation **will vary the reading on the voltmeter** (1)" "**Do not accept 'to keep the intensity of light constant'**" — that restates the variable rather than explaining it.*

Two parts: name the variable specifically, then say what it would affect and by what physical mechanism.

PART H — THE PHYSICS CONTEXTS

Unit 6 draws its contexts from **Units 1, 2, 4 and 5**. You are never asked to recall obscure theory — but you must recognise the relationship well enough to linearise it. These are the contexts that have actually appeared:

Context	Relationship used	Typical graph
Capacitor discharge	$V = V_0 e^{(-t/RC)}$	In V against t , gradient $-1/RC$
Capacitor charge sharing	$Q = CV$, conservation of charge	V_2 against V_1
Radioactive absorption	$C = C_0 e^{(-\mu x)}$	In C against x , gradient $-\mu$
Light absorption by glass	$V = Ae^{(-Bw)}$	In V against w , gradient $-B$
Air pressure with height	$P = P_0 e^{(-bh)}$	In P against h
Gas laws / absolute zero	$P \propto T$ (kelvin)	P against θ , extrapolate to P = 0
Cooling	Newton's law of cooling, exponential	$\ln(\theta - \theta_{\text{room}})$ against t
Mass–spring oscillation	$T = 2\pi\sqrt{(m/k)}$	T² against m , gradient $4\pi^2/k$
Simple pendulum	$T = 2\pi\sqrt{(l/g)}$	T² against l , gradient $4\pi^2/g$
Damped oscillation	amplitude $A = A_0 e^{(-kt)}$	In A against t
Circular motion	$F = mv^2/r$, $T = 2\pi/\omega$	F against v^2
Magnetic field of a coil	$B \propto NI/d^3$ etc.	V against $1/d^3$
EM induction / Lenz	$\varepsilon = -N d\Phi/dt$	time of fall against tube property
Standing waves	$v = f\lambda$, $f = (1/2L)\sqrt{(T/\mu)}$	f against $1/L$

Stefan / Wien (astro)	$\lambda_{\text{max}} T = 2.898 \times 10^{-3}$	peak of a curve
Luminosity–mass	$L = kM^n$	log L against log M, gradient n
Viscous flow / Stokes	$F = 6\pi\eta r v$, terminal velocity	v against r^2
Young modulus / rubber band	$E = Fl/A\Delta x$	F against Δx
Density of an object	$\rho = m/V$ with a compound shape	—
Diode / resistance with temperature	non-linear	I against θ

△ **You do not need to remember these formulae** — they are supplied or derivable from the sheet. What you need is the **linearisation habit** of §19.

PART I — THE PER-PAPER MAP

Paper	Q1	Q2	Q3 (the big one)	Q4 (measurement)
2019 June	—	Coil/induction: devise a method, graph choice, main uncertainty	Wien's law: curve, peak, temperature; criticise	Radioactive absorption: background as systematic; In C graph; gradient
2019 Oct	Density of a metal sheet	Cooling of hot water	Force between bar magnets	Damped mass–spring
2020 June	Conservation of momentum, air track, light gates	Alpha range in air — criticise a plan (6)	Bar magnet oscillation; log T vs log I	Density of a slotted mass
2021 Jan	Ball oscillating on a U-track	Capacitors in series, discharge through R	Electron diffraction	Density of a cylindrical container

2021 June	Rotating wheel + oscilloscope	Specific latent heat of vaporisation	Trolley between springs	Diffraction grating
2021 Oct	Wire-wound resistor heating	Pendulum with a peg	Standing waves	L-shaped steel rod
2022 Jan	Air column length vs temperature	Lenz's law, magnet in copper tube	Beta and gamma: precautions	Hollow cylinder rolling down a ramp
2022 June	Heating a liquid	Two coupled pendulums	Spectral lines	Metal nut: diameter, uncertainties
2022 Oct	Magnetic field of a coil	Liquid volume in a tube	Vapour pressure	Two constantan wires
2023 Jan	Bunsen flame temperature	Circular motion, rubber bung	Damping of a rotational pendulum	Metal ring: d, d², area, %U
2023 June	Water tank, constant depth	—	—	—
2023 Oct	—	Ceiling height by pendulum	$P = P_0 e^{(-bh)}$	Rubber band
2024 Jan	—	Cooling of hot water	—	—
2024 June	Electrolytic capacitor energy	Flow of water through a pipe	—	Plastic protractor: volume, %U, density deduce
2024 Oct	Diode current vs temperature	—	$L = kM^n$, log-log	Wooden metre rule: E from $16\pi^2 MD^3/ab^3 T^2$
2025 Jan	Pressure of a fixed air volume	Power input vs brightness	—	—
	Capacitor		V =	

2025 June	charging: safety, circuit, criticise graph	Absolute zero: devise a method (6), data logger	$Ae^{(-Bw)}$: ln graph, gradient, number of slides	Spring: timing, mean, %U, Poisson ratio, deduce
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The pattern is unmistakable. Question 4 is a measurement-and-uncertainty question in **every** paper. Question 3 is a linearisation-and-gradient question in almost every paper. Together they are 30 of the 50 marks.

PART J — EXAM TECHNIQUE

28. The twenty most commonly lost marks

1. **Using the full range instead of half the range** for an uncertainty — *"use of full range scores 0."*
2. **Saying "repeat and average"** without *"at different orientations / places"* — one mark instead of two.
3. **Saying an instrument is "accurate" or "precise"** instead of stating its **resolution**.
4. **Omitting the percentage-uncertainty calculation** when justifying an instrument choice.
5. **Giving a conclusion without the supporting limit** — every "deduce" mark is dependent.
6. **Missing the unit on a final answer line** — *"the final calculation mark will not be awarded."*
7. **Wrong significant figures** — gradient 2–3 s.f., %U 1–2 s.f., mean to the specified d.p.
8. **No triangle drawn on the graph** for a gradient — *"bald answer can score MP2 only."*
9. **Reading gradient coordinates from the data table** rather than from the best-fit line.
10. **Axes without units**, or a log axis labelled $\ln v$ rather than $\ln (v / \text{mV})$.
11. **A scale on multiples of 3 or 7**, or points spanning less than half an axis.
12. **Adding percentage uncertainties for a sum or difference** — add the **absolute** ones.
13. **Forgetting to multiply by the power** — D^3 means **$3 \times \%U$ in D**.

14. **Writing the log equation without comparing it to $y = mx + c$** — the second mark is dependent.
15. **"Plot a graph" without saying what against what and how it is used.**
16. **Generic safety answers** — goggles and "switch off the power" are explicitly ignored.
17. **"Eliminate random error"** — you *reduce* random error and *eliminate* systematic error.
18. **Saying "the time period will be larger"** when timing multiple oscillations — the **measured time** is larger.
19. **Using $g = 10$** — a one-mark penalty.
20. **Naming a control variable without explaining its effect** — "*to keep the intensity constant*" is rejected.

29. Priority order for revision

Priority	Content	Marks/paper
1	Combining uncertainties and the "deduce" recipe (§10–§12)	10–15
2	Linearising with ln/log and the two-line explanation (§19)	4–6
3	The plotting checklist and gradient method (§17, §18)	8
4	The six-point "devise a method" template (§21)	6
5	Instrument resolution and the two-mark justification (§5, §6)	2–4
6	The "technique to improve accuracy" pairs (§7)	2
7	Criticising a graph — the answer bank (§25)	2–3
8	Random vs systematic, data loggers (§14, §16)	2–5
9	Context-specific safety (§24)	2
10	Timing oscillations (§8)	2–3

30. The routine for every question

1. Read the **WHOLE** question, including the mark allocation and any specified significant figures or "show that" target

2. For a calculation: EQUATION → SUBSTITUTION → ANSWER → UNIT → s.f.
3. For an explanation worth 2 marks: write TWO sentences – the statement, then the consequence. Never merge them.
4. For a graph: labels with units → scale → plot to 1 mm → thin best-fit line → triangle drawn on → subtraction written out
5. For a "deduce": limit calculated → comparison stated → tentative conclusion

31. Exam-day tactics

- **You have 1.6 minutes per mark — the most generous rate in the IAL.** Plot carefully; it is worth 5 marks.
- **Use a sharp pencil and a ruler.** Points as crosses, line thin and continuous.
- **Write the unit on every answer line** before moving on.
- **Show the substitution even when you are confident** — *"use of the formula"* survives a power-of-ten slip.
- **Draw the gradient triangle large and mark its coordinates on the graph.**
- **For every two-mark explanation, write two separate sentences.** Dependency is everywhere in this paper.
- **Leave five minutes to check:** every answer has a **unit**; every gradient has a **sign and 2–3 s.f.**; every uncertainty used **half the range**; every "deduce" has a **limit and a conclusion**.